

ARHAAN GUPTA

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Education

- **Lovely Professional University**
B.Tech in Computer Science and Engineering (CGPA: 6.7)
- **Dr. Virendra Swarup Education Center**
Intermediate; Percentage: 76%

August 2023 – July 2027
Phagwara, India
April 2021 – April 2023
Kidwai Nagar, Kanpur

Experience

- **Humanbit** **May 2026 – Present**
Freelancer *Remote*
 - Annotated 1,000+ video clips across 35+ hours of focused annotation work, maintaining 98%+ labelling consistency and full adherence to quality guidelines, directly supporting downstream AI model training outcomes.
 - Coordinated a team of 8 freelancers across 3 simultaneously active projects, assigning 200+ tasks and resolving 20+ annotation queries, reducing overall turnaround time by 25%.
- **Indian Institute of Technology, Kharagpur** **June 2025 – July 2025**
Research Intern *Kharagpur, India*
 - Developed and standardized annotation protocols adopted across the research team, increasing labelling accuracy by 15% over prior baselines and ensuring high-quality data for downstream ML model training.
 - Annotated over 15,000 data samples with fine-grained Likert-scale scores, applying structured guidelines to maintain inter-annotator consistency and contributing to a 20% improvement in model performance during validation.
 - Optimized data processing pipelines for QA tasks by eliminating key sources of label noise and formatting inconsistencies, achieving a 40% decrease in error rates.

Projects

- **Sentiment Analysis: Sentiment Classification on 50K IMDB Reviews** **May 2026**
Tools: Python, scikit-learn, NLTK, TF-IDF, Streamlit, GitHub
 - Designed and developed a complete end-to-end NLP pipeline integrating TF-IDF vectorization with three classifiers (Logistic Regression, Naive Bayes, Linear SVM) benchmarked across the full 50K IMDB dataset, with the best model achieving 91.8% accuracy and an AUC-ROC of 0.963.
 - Engineered a multi-step text preprocessing module covering HTML tag removal, punctuation stripping, stopword filtering, and WordNet lemmatization, reducing vocabulary dimensionality and improving F1 score consistency across both sentiment classes.
 - Deployed the trained model as an interactive Streamlit web application hosted on Streamlit Community Cloud, version-controlled via GitHub, enabling real-time sentiment prediction with confidence scores on user-submitted reviews.
- **Medallion Architecture Data Pipeline** **April 2026**
Tools: PySpark, Spark SQL, Spark ML, Parquet, Python
 - Built a three-stage Medallion architecture pipeline (Bronze, Silver, Gold) in PySpark to ingest, clean, and transform a 12MB Amazon Cell Phones & Accessories review dataset into an ML-ready format, enforcing schema validation on ingest and partitioned Parquet storage for efficient downstream querying.
 - Engineered the Silver layer to standardize raw fields (ratings, timestamps, review text) and derive binary sentiment labels from numeric ratings, filtering null and neutral records to produce a clean, leakage-free modeling dataset.
 - Designed Gold-layer feature engineering (review length, word count) partitioned by sentiment label, then built a Spark ML pipeline (Tokenizer, StopWordsRemover, HashingTF, IDF, Logistic Regression) achieving 88% test accuracy on held-out data.

Skills

- **Programming Languages:** C, C++, Python, SQL, PL/SQL, Java, R
- **Web Technologies:** HTML, CSS, JavaScript, Node.js, React
- **Other Tools & Technologies:** Power BI, Cognos, pytsx3, OpenCV, Tkinter, scikit-learn, seaborn, pandas, NumPy, Matplotlib, Git, Linux, VS Code, PyCharm, GitHub, Microsoft Excel, MongoDB
- **Research Skills:** Literature Review, Model Evaluation, Data Collection, Monitoring, Preprocessing

Certifications

- [Marketing Analytics \(Meta\)](#) June 2026
- [Cloud Analytics \(Oracle\)](#) October 2025
- [R Programming for Business Analytics \(NPTEL\)](#) October 2025
- [Relational Database Management System \(IBM\)](#) March 2025
- [Python \(IBM\)](#) March 2025
- [Software Development Processes and Methodologies \(Coursera\)](#) March 2024

Achievements

- [Successfully completed Deloitte's Data Analytics Job Simulation.](#)
- Achieved Rank 4417 globally in the Master of All category in Mobile Legends: Bang Bang (MLBB).
- Finalist in the Smart India Hackathon (Software).
- Ranked among the top 10 teams in the MEDHA Hackathon.
- Secured 2nd place in the Celestia and Code Caravan hackathons.
- Coordinated marketing operations as part of the Sunburn event team.